CSCI2160Quiz2F11BinaryHex Name: Brandon Buchanan

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1. (21 pts)

Character	Hollerith	EBCDIC binary	EBCDIC hex	ASCII binary	ASCII hex	ASCII Decimal
F	12-6	1100 0110	C6	0100 0110	~~47h~~ 46	~~71~~ 70
J=11-1 N	11-5	1110 0101	D ~~E~~5	0100 0101	~~55h~~ 4E	~~85~~ 78
b	Do not	Answer	These 3	0110 0010	62h	98
7	7	1111 0111	F7	0011 0111	37h	55

-7

I = 50h
A = 41h
a = 61h

2. (20) Perform the below additions in the appropriate bases

$$\begin{array}{r} 11011111_2 \\ \underline{01101111_2} \\ 101001110_2 \end{array}$$

$$\begin{array}{r} 2EB2E_{16} \\ \underline{FC28F_{16}} \\ 12AFBDh \end{array}$$

D

-5

3. (30 pts) Represent the below integers as 2-byte binary and hex values. Remember that the representation MUST be in 2 bytes. Also, be sure to use 2s complement for negative numbers.

	Binary	Hex
43	00000000 00101011	002B
-9	11111111 11110111	FFF7
-127	1111 1111 1000 0001	FF81

00h 00001001
FFh 11110110 $\xrightarrow{+1}$ FFF7h

00h 01111111
FFh 10000000 +1 = FF81h

4. (29) Given the hex value 0E8h (Note that the 0 is required in assembly so that the assembler does not assume that this is a variable. This is a simple 2-hex digit value), interpret the byte as an

8-bit signed decimal value: -24

E8h $\xrightarrow{flip}$ 17h $\xrightarrow{+1}$ 18h → -24d

8-bit unsigned decimal value: 232

16 ×14 160 + 64 224 + 8

An EBCDIC character: Q ~~X~~ Y

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